



Battery recycling

Expectations in Indian markets

Apart from EV, there are applications like robotics coming up in the forefront in a big way. From a user's point of view, there are certain expectations from the battery manufacturers.

Talking about battery usage in robotics and other systems, Kumar finds more lithium-ion batteries, lithium-iron phosphate (LFP) batteries, and lithium-titanate oxide (LTO) batteries in action. He explains, "The number one requirement as a robotics supplier is safety. I don't want my robot to harm anyone or any other machine, or harm itself, by the kind of safety system in the battery or the battery management system (BMS). The other things are efficiency and fast charging as we have customers asking the number of hours it can run and if the fast charging is limited to 1C, 3C, or 5C?"

While experts discuss on the ground challenges, there is also a concern on temperature and its control techniques. Kumar says, the BMS is very important, but as a robot builder, I should not be building BMS. He thinks, "Application decides BMS, and it needs more understanding, studies, and rewriting algorithms. It has to be according to

the application which will suit my needs."

Another important expectation he points out is Wi-Fi charging ability, which is now offered by many of the European competitors.

Decoding battery chemistries

There is often a confusion on what will be the right chemistry for the battery to make it safe or long-lasting. There is also an ongoing hype about sodium batteries that are heavier and

cheaper than lithium-ion. Some serious efforts in working under the skin of batteries has led to more innovation in recent years. So, here are a few things to consider.

Chemistry depends on product design. The technological first style lithium-ion chemistry is still found to be interesting by industry experts due to its variations having different types of command cells and also other chemistries like LFP, Ca doped LTO, and sodium ion cells. Solid-state batteries are expected to grow as the energy sector is rapidly innovating and trying to build products that can be commercialised for a long time.

Naskar says, "As a technology developer, I can tell there is no specific one answer to the question of which chemistry would really be suited for a long term. It really depends on your product design and the kind of market you want to deliver to. For some, safety and light weight will matter and for others it may not."

Most suited chemistry for India.

Naskar strongly feels LFP will be a way forward specifically for the tropical climate of India. He says, "After testing multiple chemistries by benching them against each other, and considering our climate, road conditions, and the driving cy-

cle, if we go for an LFP battery pack, it provides the safety we want from the system."

Although this decision may take a hit on the mass budget for the battery pack, the overall cost of the system comes down, meeting our functional requirements on the vehicle. Naskar thinks this may not be true for a two-wheeler manufacturer as the vehicle itself is very lightweight and would need a much lighter system. He says, "It depends on what type of system you are building."

Patwardhan also believes that LFP is the right chemistry for India. So, when you look at the picture in NMC (lithium nickel manganese cobalt oxide), chemistry is appropriate. Ultimately, any chemistry is appropriate, but the question is how effectively you manage the spiral chart. If you look at all the chemistries, they have a different axis—power, density, safety, and many standards which just cannot be compromised. He explains, "For an LFP battery, we take a drop test multiple times from 150ft and have it crushed with a bulldozer just to demonstrate the appropriateness of a chemistry needed for the Indian conditions that are abusive in terms of environment."

Sodium-ion: The future. The more recent leg in the battery chemistry race is sodium-ion combination. Done right, this technology could lead to widespread adoption in the market. Experts believe this new chemistry has a future because of limited resources available for predecessors like lithium-ion for which India will continue to be dependent on the global supply-chain, even if demands of manufacturing are internalised.

Being a peninsular country with coastlines, sodium is more abundantly available in India. Companies like Faradion are already trying to set up a sodium-ion based cell manufacturing facility in India. The industry experts opine this would be the next big thing.

"It is probably in the lab right